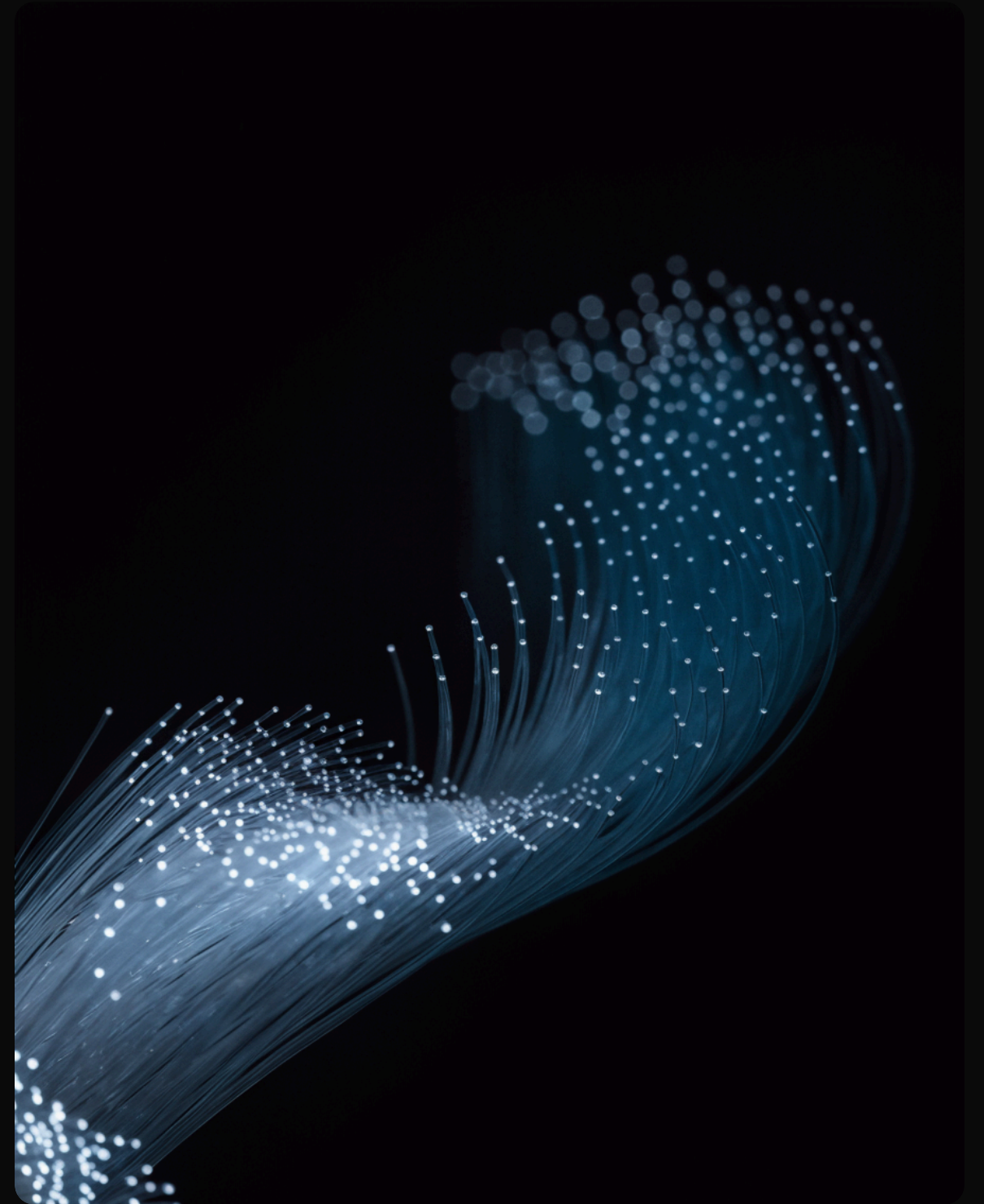


IDS Project

Phase 2: Deep sequential anomaly detection

Adaptive Cyber-Physical Security





Phase 1 limitations: where GMM fell short

01

Bot — 45.2% F1

C2 beaconing relies on temporal patterns across flows. GMM sees each flow in isolation and misses the sequence.

02

Infiltration — 33.1% F1

Multi-stage attack with only 36 flows. Too few samples and requires cross-flow context to detect.

03

Web Attack XSS — 3.1% F1

Low-volume, flow-level features are indistinguishable from normal traffic without sequential context.

04

DoS GoldenEye — 54.8% F1

Traffic patterns overlap with normal activity. Temporal sequencing is needed to reveal the attack signature.

01 Train on normal only

Model learns exclusively from benign traffic. Attacks are never seen during training — making detection label-free.

02 Reconstruction error

Traffic flows through Encoder → 32-dim latent space → Decoder. Normal sequences reconstruct cleanly; attacks produce high error and trigger an alert.

03 LSTM autoencoder

Recurrent architecture. Memory cells capture short-range sequential dependencies across the traffic sequence.

04 Transformer autoencoder

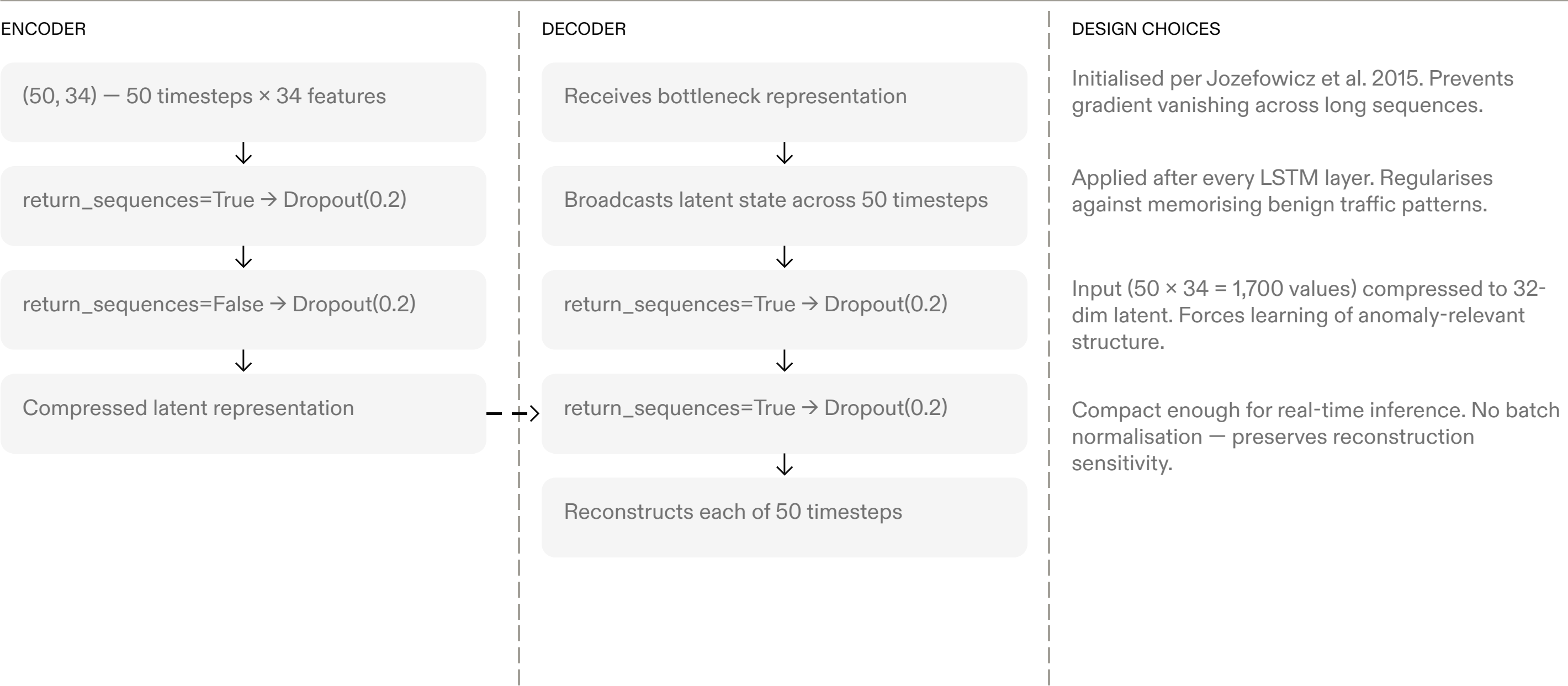
Attention-based architecture. Captures global temporal correlations across the full sequence simultaneously.

Sequence construction

WINDOW	Window size $W = 50$ Covers C2 beacon intervals (30–300 s) and multi-stage attack sequences	~5 min
STRIDE	Training vs. evaluation stride Stride 10 for training efficiency; stride 1 for full evaluation coverage	10 / 1
WINDOWS	Training / evaluation windows 121,464 training windows (0.83 GB) · 716,043 evaluation windows (4.87 GB)	716K
FEATURES	Feature dimensions = 34 Post RobustScaler normalisation and correlation-based feature removal	34

LSTM-AE architecture

262,978 parameters · 53× compression bottleneck



LSTM gate equations

Forget gate — what to erase from memory

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Input gate — what new information to store

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

Cell update — actual memory update

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Output gate — what to pass forward

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(c_t)$$

Why LSTM over vanilla RNN? Gating prevents vanishing gradient across $W=50$ timesteps.

Transformer-AE architecture

Attention(Q,K,V) = softmax(QKT / $\sqrt{d_k}$) V
Input (50,34) → Dense(64) → Sinusoidal PE → 2× TransformerBlock → GAP → Dense(32) → RepeatVector(50) → 2× TransformerBlock → TimeDistributed(Dense(34))

Criteria	LSTM-AE (Model B)	Transformer-AE
Inductive bias	Sequential recurrence	Global attention
Positional info	~ Implicit (recurrence)	✦ Sinusoidal PE (explicit)
Parameters	262,978 more params	167,906 fewer params
Selected	✓ Yes Model B	✗ No Not selected

Training protocol

Loss function

MSE reconstruction loss averaged over all timesteps and features across N windows.

Optimiser

Adam, lr=0.001, gradient clipping (max norm=1.0)

Callbacks

- ModelCheckpoint — saves best val_loss epoch
- ReduceLROnPlateau — halves lr after 5 plateau epochs (min=1e-6)

Anomaly score per window

Average squared reconstruction error across all timesteps and features.

Threshold selection

τ = Percentile95 of anomaly scores on the benign validation set, allowing a 5% false positive rate on benign traffic.

Ablation study

4 controlled variants to isolate each design decision.
Baseline: Full LSTM-AE with dropout, W=20, no PE, latent d=32.
Results pending run_ablation.py.

Variant	What's removed	Expected impact
No dropout	Dropout(0.2) removed	↑ overfit to benign dist.
Short window (W=10)	W=20 → W=10 halved context	↓ recall on multi-step attacks
No positional encoding	PE removed Transformer only	↓ order sensitivity in attn.
Small latent (d=8)	d=32 → d=8 tighter bottleneck	↓ temporal structure preserved
Why these 4?	Regularisation	Context · Bias · Bottleneck
Status	Pending	run_ablation.py

LSTM-AE results

100%

Recall — all 14 attack types

63.94%

F1-Score

47.0%

Precision — threshold artefact

52.19%

AUC-ROC — shuffle artefact

0.031

KS stat — near-zero separation

0.83 ms/seq

Inference — CPU, real-time

AUC $\approx$ 52% and Precision $\approx$ 47% reflect shuffle artefact (shuffle=True mixes $\sim$ 47% attacks per window). Phase 3 fix: temporally ordered evaluation. Bot: 45% $\rightarrow$ 100%, Infiltration: 33% $\rightarrow$ 100% vs GMM.

Conclusion & next steps

Phase 2 achievements

Two sequential architectures trained and compared: LSTM-AE and Transformer-AE. 100% recall across all 14 CICIDS-2017 attack types. Bot and Infiltration detection improved from 45% → 100% over Phase 1 GMM. 4-variant ablation study quantified each design choice. 0.83 ms/sequence CPU inference confirms real-time feasibility.

Current limitation

Shuffled test set makes AUC unreliable — the 52.19% score is an evaluation artefact, not a true performance signal. Temporally ordered evaluation is required.

Phase 3 — Model C (Hybrid Ensemble)

Hybrid score fusion combining GMM (per-flow discrimination) and LSTM-AE (temporal pattern detection), weighted by α . Temporally ordered evaluation with concept drift simulation: train Monday, test Friday. Expected gains: Bot and Infiltration precision from Phase 2 combined with GMM precision from Phase 1.

Phase 3 accounts for 40% of the final grade.